Cisco Virtual Internship Programme 2025

Internship Documentation

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Course: B.E. – Computer Science And Engineering (Vth Semester)
College: Methodist College of Engineering & Technology, Hyderabad

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Under the Guidance of:
Cisco Networking Academy (NetAcad)
In collaboration with
Department of Computer Science And Engineering,
Methodist College of Engineering & Technology

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Problem Statement:-

PART 1:

Problem Statement: You are a part of the cybersecurity student team at your college, freshly enrolled in the Cisco NetAcad Cybersecurity course. With access to Cisco Packet Tracer and your growing knowledge of security fundamentals, you've been given your first real-world challenge. Your task is to analyze your own college network as if you were part of an internal red team.

You'll begin by mapping the current infrastructure using Cisco Packet Tracer, identifying devices, access points, firewalls, segmentation boundaries, and any existing security controls. But this isn't just a drawing exercise. You are expected to assess how effective these controls are in today's threat landscape. Where are the weak points? Are there flat zones that allow lateral movement? What would an attacker target first, and how would you stop them? Using the knowledge from your NetAcad course and insights gained through simulation, conduct an attack surface analysis, and present your findings. Your recommendations should reflect real-world thinking, assume budgets are tight, staff are limited, and security is everyone's afterthought until something breaks.

Tasks:

● Conduct a complete analysis of the existing college campus network layout, devices, and zones.

● Use Cisco Packet Tracer to create a visual representation of routers, switches, firewalls, and

access points.

● Assess how the network is segmented and which trust zones exist.

● Identify and document any security controls such as firewalls, IDS/IPS, authentication servers, or

ACLs.

● Perform an attack surface mapping exercise to locate potential weaknesses.

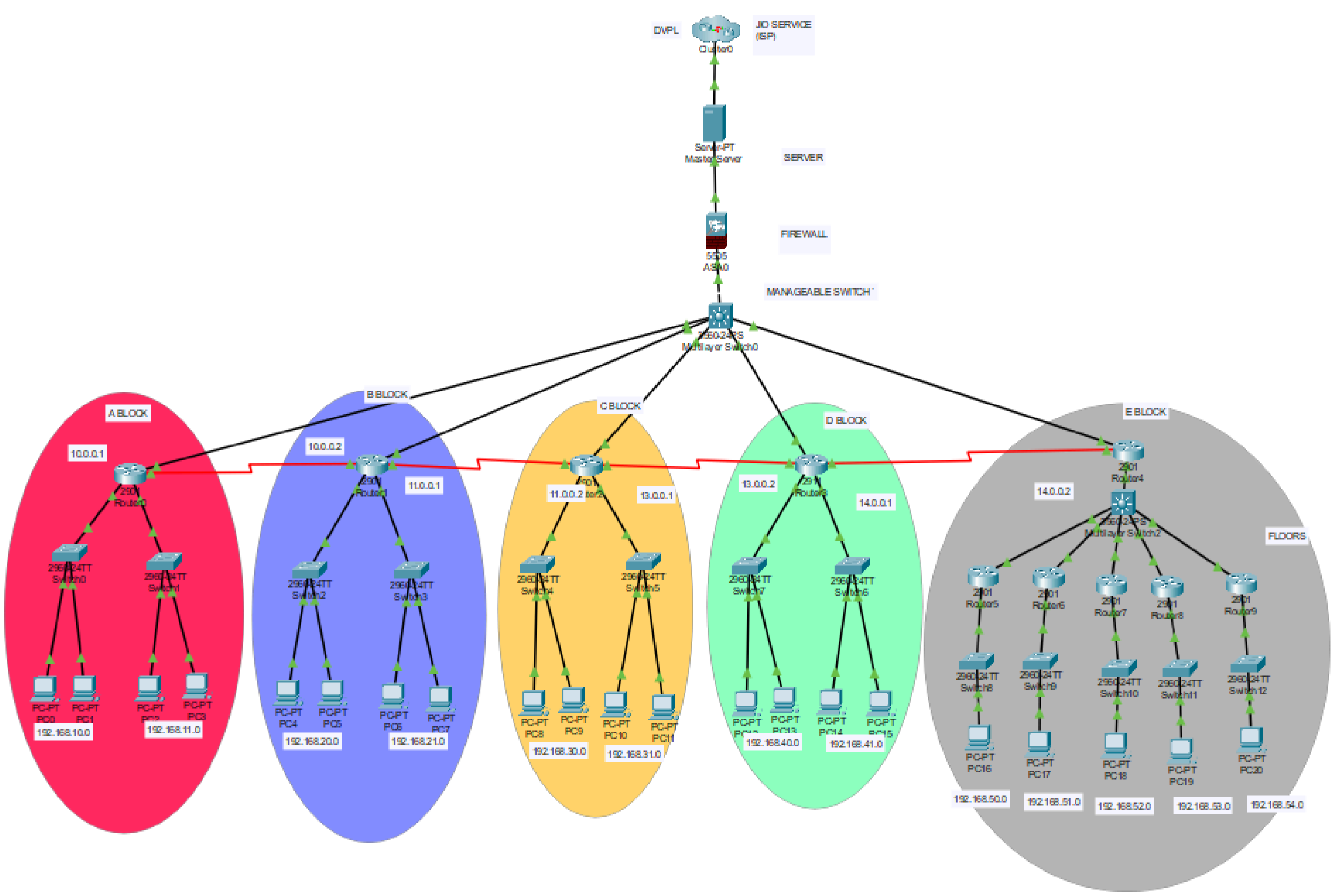
● Suggest risk-based countermeasures, policy changes, and improved control placement

Solution:-

The college network begins with two internet service provider connections, which are routed into the servers and then pass through a firewall for initial security filtering. From the firewall, connectivity flows into a central manageable switch that acts as the backbone of the campus. This switch connects to routers installed in each block, and these routers distribute the connection to departmental switches, which in turn connect to offices, staff rooms, labs, and WiFi access points. This layered design ensures that every academic and administrative area has access to both wired Ethernet and wireless internet services.

Network Design Framework

This is the overall architecture of our college, divided into 5 distinct blocks



The college consists of five distinct blocks:-

A Block (Administration & H&S Department)

Examination department:-

- A central switch, connected to 5 PC's
- A dedicated server to store results and student records
- Printers and VOIP's for communication

Admission cell:-

- VOIP's,3 PC's and printers

Director room:-

- Wireless router, 2 pc's, 1 printer

Cash office:-

- 5 pc's and a printer

Board room:-

- 1 projector and 3 pc's

Accounts department:-

- Accounts related to the faculties and students

Grievance cell:-

- Various grievances from the faculty and students

Classrooms (5):-

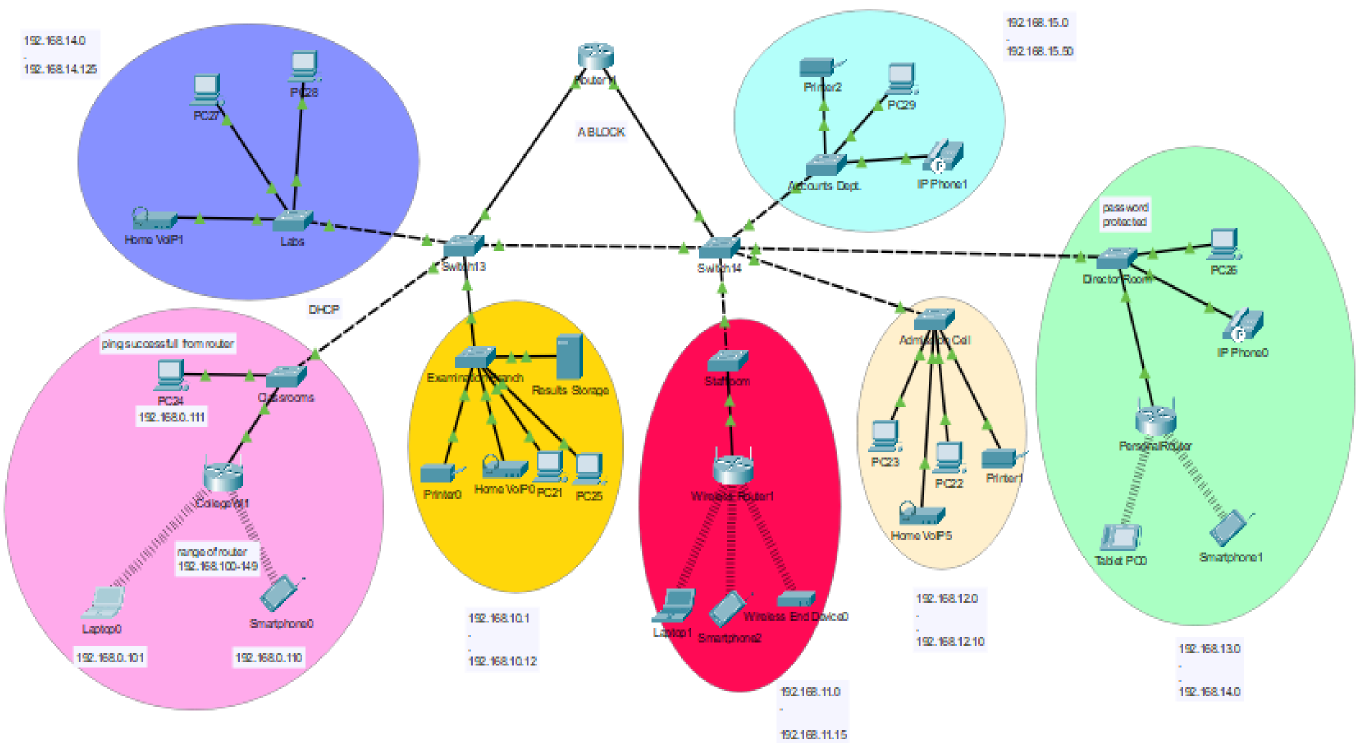
- 1 PC for each class
- 1 wireless projector

Labs:-

- 40 PC's each and 1 projector

Loopholes:-

The weak point in Block A is that critical administrative departments such as Accounts and Examination share the same network with student labs and staff systems. This lack of segmentation exposes confidential academic and financial data to potential misuse, unauthorized access, and accidental leakage.



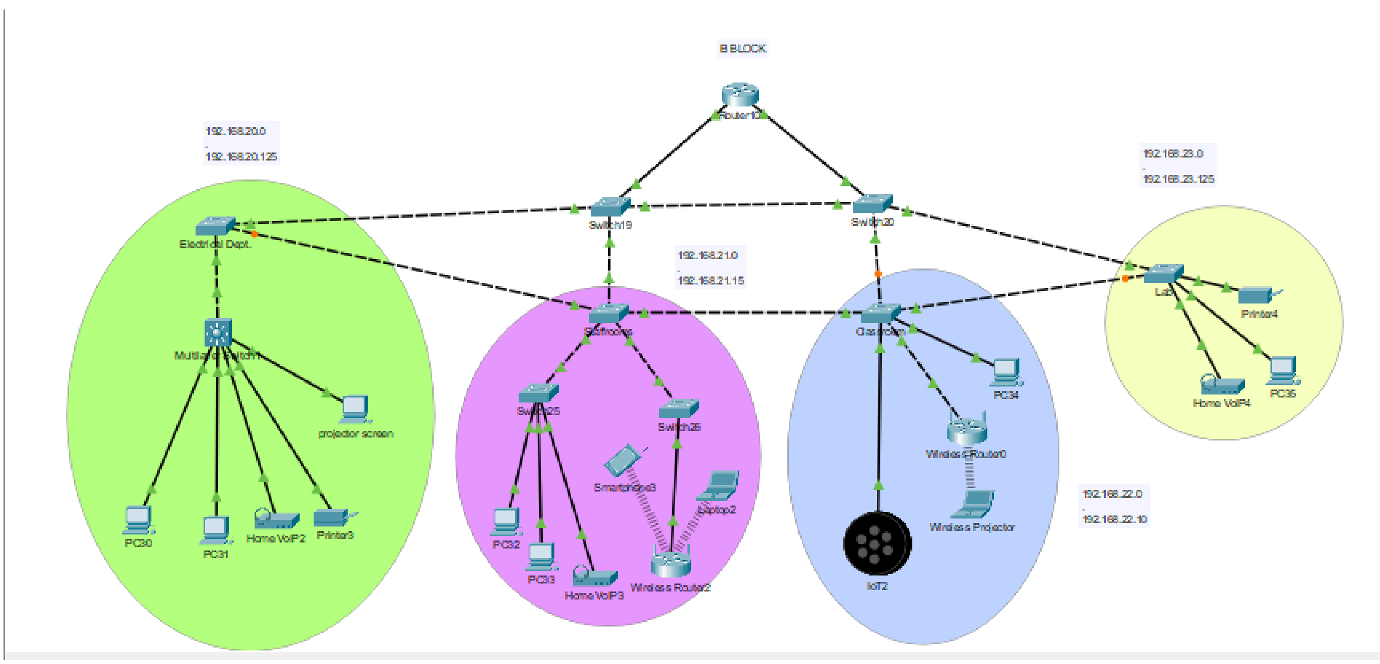
B Block (Electrical & ECE Block)

Electrical department:-

- 10 Pc's
- Staff rooms
- Classrooms
- Labs
- Hardware components

Loopholes:-

The weak point in Block B is the absence of separation between staff and student systems. Since both groups use the same network segment, there is a risk of student traffic interfering with staff operations. High usage in labs could also affect staff productivity due to bandwidth sharing.



C Block (Mechanical/Civil Block)

Labs:-

- 2 labs, 40 systems each on 1 lab

Staffroom:-

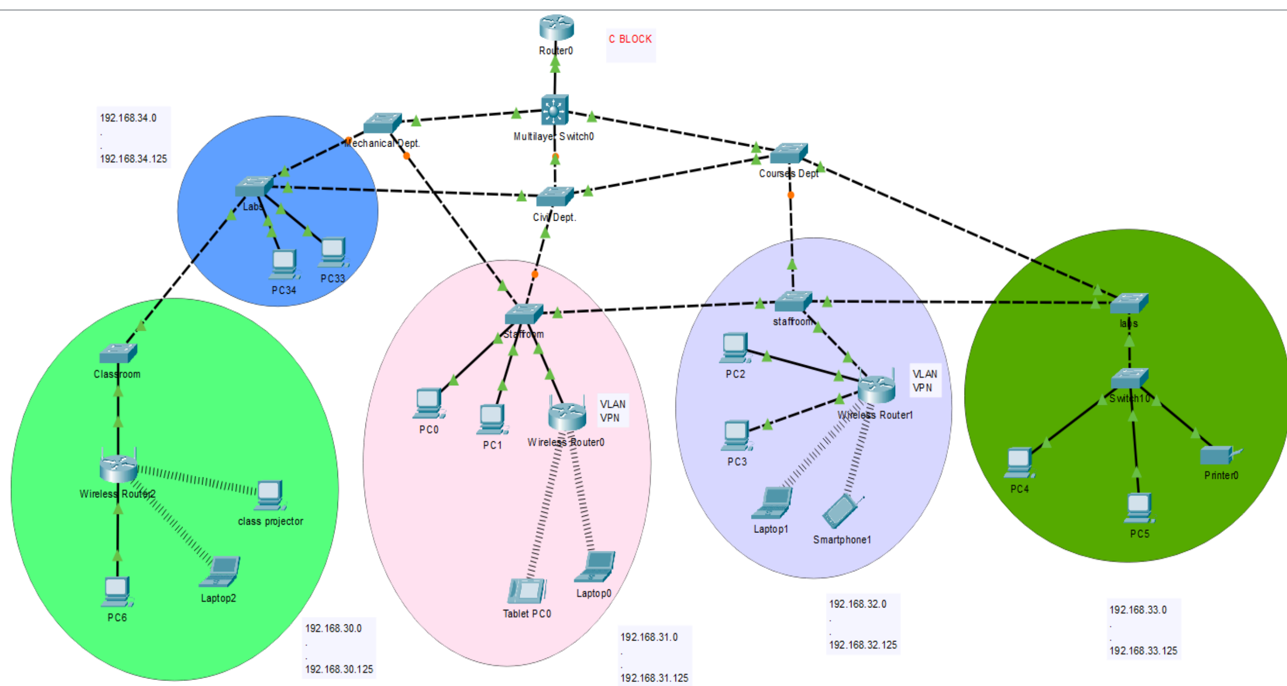
- On every floor
- 10 Pc's

Classrooms:-

- · Civil/mechanical departments

Loopholes:-

The weak point in Block C lies in its flat network design, where lab systems and faculty systems share resources without logical isolation. If malware spreads through student labs, it can potentially impact faculty systems, and unauthorized access attempts may go undetected in such an open structure.



D Block (Library & Seminar Hall)

Library

- 5 PCs
- Dedicated server for library books

Digital Library

- For digital learning
- 20 PCs

Classrooms

- 2 classrooms, 1 projector in each class

Seminar Hall

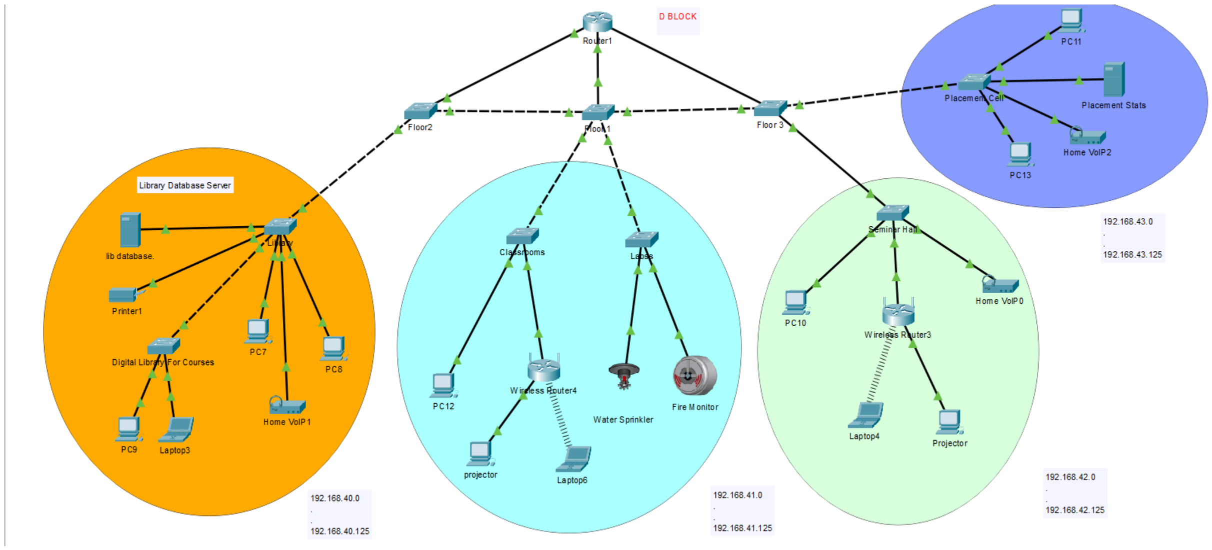
- 2 PCs and 1 projector

Labs

- Hardware labs
- IT workshop
- Hardware components like CPU, Mouse, Storage Devices, etc

Loopholes:-

The weak point in Block D is that the library and digital library, which store and deliver academic resources, are not secured separately and are on the same segment as the seminar hall. Since the seminar hall often accommodates guest users during events, this creates a serious risk of unauthorized access to digital resources and possible misuse of academic bandwidth.



E Block (CSE Block)

- 5 floors
- A central switch and router on each floor

Labs

- 35 PCs on each floor
- 1 wireless projector on each floor

Soil Engineering Department

- 1 PC and 1 wireless projector

Networks Department

- 2 PCs and 1 printer
- Wireless router

Staff Rooms

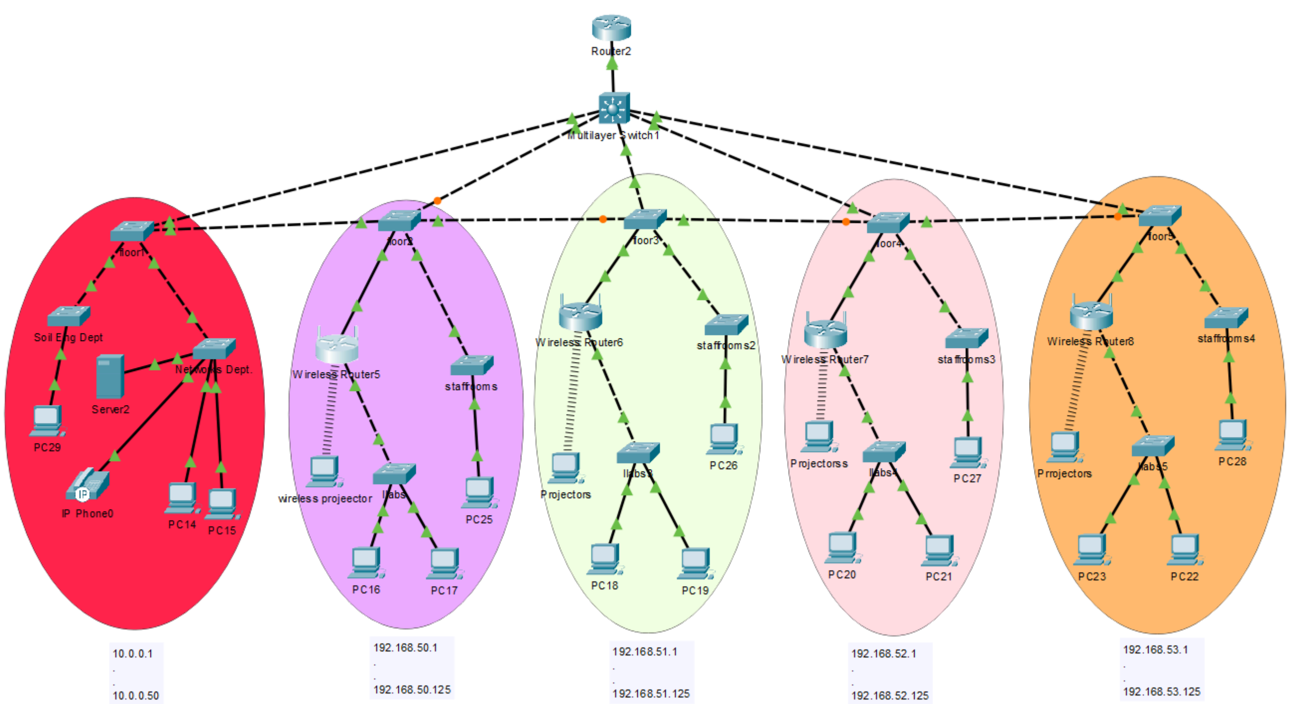
- 10 PCs on each floor

Classrooms

- 1 PC and 1 wireless projector on each floor

Loopholes

The weak point in Block E is the very high density of systems and labs connected within a single structure. Without VLAN segmentation, there is a strong chance of bandwidth bottlenecks, malware propagation, and unauthorized lateral movement across labs and staff networks. Furthermore, the presence of WiFi throughout the block increases the attack surface, as unsecured or guest devices could connect to sensitive networks if not properly isolated.



PART-2

Problem Statement: After your impressive audit in Part 1, the college IT department has invited you to contribute to a new project: enabling a hybrid access model for students and faculty. Faculty members will now work flexibly from home or campus, and require uninterrupted, secure access to teaching tools, research repositories, and internal services. Students, on the other hand, will continue using personal devices to access shared academic portals and lab resources. But here's the catch: the administration has made it clear that the internal services must never be exposed directly to the internet. Your task is to design a secure hybrid network architecture that supports remote access while enforcing strict boundaries. Think like a network engineer and evaluate options like VPN, SASE, identity-aware proxies, or split tunneling. Consider not only how to connect, but how to ensure the right people access the right services at the right time. Can your design balance simplicity, security, and scale without overwhelming the existing infrastructure?

Tasks & Deliverables:

● Design network segmentation based on user roles (faculty vs student).

● Recommend secure access tools like VPN, SASE, identity-aware proxies, or split tunnelling.

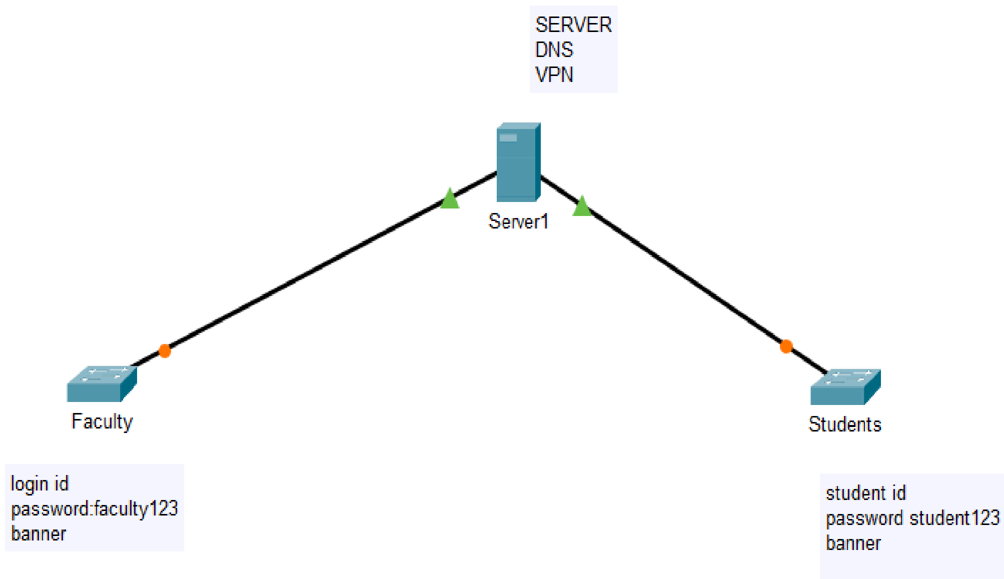
● Define trust models, authentication flows, and control access to internal apps.

● Update the campus network topology to show remote access pathways, gateways, and policy

enforcement zones.

● Justify your architecture with risks, use cases, and fallback strategies.

Solution:-



2.0 Hierarchical Network Model Implementation

The college network follows the **three-tier hierarchical model** designed for scalability and reliability:

Core Layer

- High-speed backbone connecting all five blocks
- Redundant links and internet gateway
- Primary function: Fast packet switching

Distribution Layer

- Block-level aggregation switches
- Inter-VLAN routing and policy enforcement
- WAN access and filtering

Access Layer

- End-user connectivity within each block

- Access switches and wireless access points
- Direct connection to end devices

2.1 VLAN Design Strategy

VLAN ID	Name	Purpose	Assigned Blocks
VLAN 10	Management	Network device management	All blocks
VLAN 20	Administrative	Administrative functions	A Block
VLAN 30	Academic	Academic departments	B, C Blocks
VLAN 40	Students	Student services	D Block
VLAN 50	Technical	Technical departments	E Block
VLAN 60	Servers	Server infrastructure	Server room
VLAN 99	Guest	Guest access	All blocks

3. IP Addressing Scheme

3.1 Subnet Allocation

Block	Network	Subnet Mask	Default Gateway	DHCP Range	Purpose
A Block	192.168.10.0/24	255.255.255.0	192.168.10.1	192.168.10.10-200	Administrative
B Block	192.168.20.0/24	255.255.255.0	192.168.20.1	192.168.20.10-200	MBA Academic
C Block	192.168.30.0/24	255.255.255.0	192.168.30.1	192.168.30.10-200	Mech/Civil Academic
D Block	192.168.40.0/24	255.255.255.0	192.168.40.1	192.168.40.10-200	1st Year Students
E Block	192.168.50.0/24	255.255.255.0	192.168.50.1	192.168.50.10-200	CSE Technical
Management	192.168.100.0/24	255.255.255.0	192.168.100.1	Static assignment	Network Management
Servers	192.168.200.0/24	255.255.255.0	192.168.200.1	Static assignment	Servers/Services

3.2 Static IP Reservations

Device Type	IP Address	Purpose
Core Router	192.168.100.1	Primary gateway
DNS Server	192.168.200.10	Domain name resolution
DHCP Server	192.168.200.11	IP address assignment
Web Server	192.168.200.12	College portal
Email Server	192.168.200.13	Email services
FTP Server	192.168.200.14	File transfer

4. Packet Tracer Implementation

4.1 Device Selection

Core Layer Devices

- **Router**: Cisco ISR 2811/2901
- **Core Switch**: Cisco Catalyst 3650/3850

Distribution Layer Devices

- **Distribution Switches**: Cisco Catalyst 3650/2960 (one per block)

Access Layer Devices

- **Access Switches**: Cisco Catalyst 2950/2960
- **Wireless Access Points**: Cisco Aironet series
- **End Devices**: PCs, laptops, printers, servers

4.2 Physical Topology Configuration

Step 1: Core Infrastructure Setup

1. Place core router in central location
2. Connect core router to internet cloud
3. Add core switches for redundancy
4. Establish high-speed links between core devices

Step 2: Distribution Layer Implementation

1. Add one distribution switch per block
2. Connect each distribution switch to core layer
3. Configure trunk links for VLAN traffic
4. Implement spanning tree protocol for loop prevention

Step 3: Access Layer Deployment

1. Add access switches for each department/floor
2. Connect access switches to distribution switches
3. Configure access ports for end devices
4. Deploy wireless access points as needed

4.3 VLAN Configuration

Switch Configuration Commands

```
Switch(config)# vlan 10
Switch(config-vlan)# name Management
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name Administrative
Switch(config-vlan)# exit

Switch(config)# vlan 30
Switch(config-vlan)# name Academic
Switch(config-vlan)# exit

Switch(config)# vlan 40
Switch(config-vlan)# name Students
Switch(config-vlan)# exit

Switch(config)# vlan 50
Switch(config-vlan)# name Technical
Switch(config-vlan)# exit

Switch(config)# vlan 60
Switch(config-vlan)# name Servers
Switch(config-vlan)# exit

Switch(config)# vlan 99
Switch(config-vlan)# name Guest
Switch(config-vlan)# exit
```

Trunk Port Configuration

```
Switch(config)# interface fastethernet 0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan all
Switch(config-if)# exit
```

Access Port Configuration

```
Switch(config)# interface fastethernet 0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
```

4.4 Inter-VLAN Routing Configuration

Router-on-a-Stick Configuration

```
Router(config)# interface fastethernet 0/0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface fastethernet 0/0.20
Router(config-subif)# encapsulation dot1q 20
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit

Router(config)# interface fastethernet 0/0.30
Router(config-subif)# encapsulation dot1q 30
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
```

4.5 DHCP Configuration

```
Router(config)# ip dhcp pool A_BLOCK
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 192.168.200.10
Router(dhcp-config)# exit

Router(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.9
```

PART-3

Problem Statement: Soon after the hybrid model rolls out, complaints start coming in: students are streaming videos during lectures, torrenting files in labs, and bypassing basic restrictions using browser extensions and proxies. The administration turns to you again, and this time for a solution that restricts web access smartly, without creating backlash or blocking legitimate research. You must design a policy framework that considers:

● Who the user is (student, faculty, guest)

● When they're online (class hours, weekends)

● What content they're trying to access (education, social media, games, etc.)

Explore modern filtering tools: DNS-based filtering, L7 firewalls, proxies, and endpoint-based

enforcement. Draft simple, understandable rules, but back them with solid policy logic and enforcement mechanisms. Don't just stop at blocking sites but instead log events, anticipate circumvention attempts, and define how violations should be reported.

Tasks:

● Compare filtering solutions: DNS filtering, Layer 7 firewalls, proxy-based, or client-side

enforcement.

● Design policies that vary by user groups, access time, or category.

● Simulate the enforcement using simple commands or pseudo-policies.

● Add components to the network that enforce and monitor these rules.

● Plan logging and alerting for any access violations

Solution:-

5. Network Services Configuration

5.1 DNS Server Setup

- Configure DNS server with college domain
- Create records for all network services
- Configure reverse DNS lookup

5.2 Web Server Configuration

- Deploy college portal website
- Configure virtual hosts for different departments
- Implement SSL certificates for security

5.3 Email Server Setup

- Configure SMTP/POP3/IMAP services
- Create email accounts for staff and students
- Implement email security policies

5.4 FTP Server Configuration

- Set up file sharing services
- Create department-specific directories
- Configure user access permissions

6. Security Implementation

Policy Framework

User Role	Access Allowed	Access Denied	Notes
Students	Academic portals, research	YouTube, Netflix, Torrents, Social Media	Restrictions during class hours
Faculty	Full access for research	Malware/adult sites	Flexible access needed
Admin	Full unrestricted	None	Administration work requires full access

6.1 Access Control Lists (ACLs)

```
Router(config)# access-list 100 permit tcp 192.168.10.0 0.0.0.255 any eq 80
Router(config)# access-list 100 permit tcp 192.168.10.0 0.0.0.255 any eq 443
Router(config)# access-list 100 deny ip any any
Router(config)# interface fastethernet 0/0.20
Router(config-subif)# ip access-group 100 in
```

6.2 Port Security Configuration

```
Switch(config)# interface fastethernet 0/2
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum 2
Switch(config-if)# switchport port-security violation shutdown
Switch(config-if)# switchport port-security mac-address sticky
```

6.3 Wireless Security

- Configure WPA2-Enterprise authentication
- Implement RADIUS server for user authentication
- Set up guest network isolation

7. Testing and Verification

7.1 Connectivity Tests

Basic Ping Tests

1. Test connectivity within same VLAN
2. Test inter-VLAN communication
3. Test internet connectivity
4. Test DNS resolution

Service Accessibility Tests

1. Verify web server access from all VLANs
2. Test email server functionality
3. Validate FTP server access
4. Check DHCP lease assignment

7.2 Performance Verification

- Monitor bandwidth utilization
- Test network latency between blocks
- Verify redundancy and failover
- Validate QoS implementation

7.3 Security Testing

- Test ACL effectiveness
- Verify port security enforcement
- Validate wireless security
- Check firewall rules

8. Documentation Requirements

8.1 Network Diagrams

- **Physical Topology**: Shows actual device placement and cabling
- **Logical Topology**: Illustrates data flow and VLAN structure
- **IP Addressing Diagram**: Details subnet allocation and routing

8.2 Device Inventory

Device Type	Model	Quantity	Location	IP Address	Purpose
Core Router	Cisco 2811	1	Server Room	192.168.100.1	Inter-block routing
Core Switch	Catalyst 3650	1	Server Room	192.168.100.2	Core switching
Distribution Switch	Catalyst 3650	5	Each Block	192.168.x.253	Block aggregation
Access Switch	Catalyst 2960	15+	Departments	192.168.x.252	End-user access
Wireless AP	Aironet 1815i	20+	All floors	DHCP assigned	Wireless access

8.3 Configuration Backup

- Complete router configurations
- Switch startup configurations
- VLAN database exports
- Security policy documentation

9. Maintenance and Monitoring

9.1 Regular Maintenance Tasks

- Weekly configuration backups
- Monthly performance reviews
- Quarterly security audits
- Annual hardware inventory

9.2 Monitoring Tools

- Network monitoring software
- Bandwidth utilization reports
- Security event logging
- Performance metrics dashboard

10. Future Expansion Planning

10.1 Scalability Considerations

- Additional VLAN capacity
- Bandwidth upgrade paths
- New block integration
- Technology refresh cycles

10.2 Emerging Technologies

- IPv6 implementation roadmap
- Software-defined networking (SDN)
- Cloud integration planning

- IoT device support

11. Conclusion

This comprehensive network design provides a robust, scalable, and secure infrastructure for the college's five-block campus. The hierarchical design ensures optimal performance while maintaining security and manageability. The documentation meets all requirements for the Cisco Virtual Internship 2025 program and serves as a complete reference for network implementation and maintenance.

12. Appendices

Appendix A: Configuration Templates

- Router configuration template
- Switch configuration template
- VLAN configuration checklist
- Security configuration guide

Appendix B: Troubleshooting Guide

- Common connectivity issues
- VLAN troubleshooting steps
- Performance optimization tips
- Security incident response

Appendix C: Vendor Information

- Cisco device specifications
- Recommended hardware models
- Software version requirements
- Support contact information

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